

Does losing a game of chess make you lose the next one?

What 87m games of online chess say about tilt: how much a loss costs, why, and what players do next

Yes, a little. In 87m games of online chess, players who have just lost move faster, blunder more and score worse in the next game. But half of what chess players call "tilt" is only a bad evening

Lichess open database: 87.3m rated bullet, blitz and rapid games, September 2026; August 2026 used to repeat the first version

1.3

Percentage points of score that a loss, instead of a win, costs in the next game. About what a player loses by dropping 14 rating points

2.8

The gap between players who lost and players who won, before form is taken out. Twice as large

7%

Less time spent on the early moves of the next game after a loss. Blunders rise by 4%

79%

Share of players who start another game after a third loss in a row. After a first loss it is 65%

Poker players call it tilt. A bad beat rattles you, you play the next hand worse, and the losses pile up. Chess players borrowed the word, and many believe in the thing. The usual advice is to stop for the night after two or three losses in a row.

Online chess lets the belief be tested. Lichess, a free chess site, publishes every rated game played on it. We took all 87m bullet, blitz and rapid games from September 2026, played from 1.9m accounts, put each player's games in order, and looked at what follows a loss.

What it looks like

At first sight, tilt is everywhere. A player's rating predicts how often they should win against any opponent, so every result can be set against that prediction. Players who have just won score 1.2 percentage points more than predicted in their next game of the same sitting. Players who have just lost score 0.4 points less, even though their rating has already been cut for the loss. The longer the run, the worse it gets. After five or more losses in a row players score 3 points less than predicted, and in rapid games nearly 8.

Losing looks catching

Score compared with what the two ratings predict, percentage points, by the number of losses in a row just before the game



Sources: Lichess open database, rated bullet, blitz and rapid games, September 2026; Signum Average calculations
A win scores 1, a draw ½. The rating is the one shown at the start of the game

None of this shows that losing does any harm. A player who is tired, distracted or drunk loses one game and then the next, with no help from the first loss. Bad form that lasts an evening looks, in these numbers, exactly like tilt.

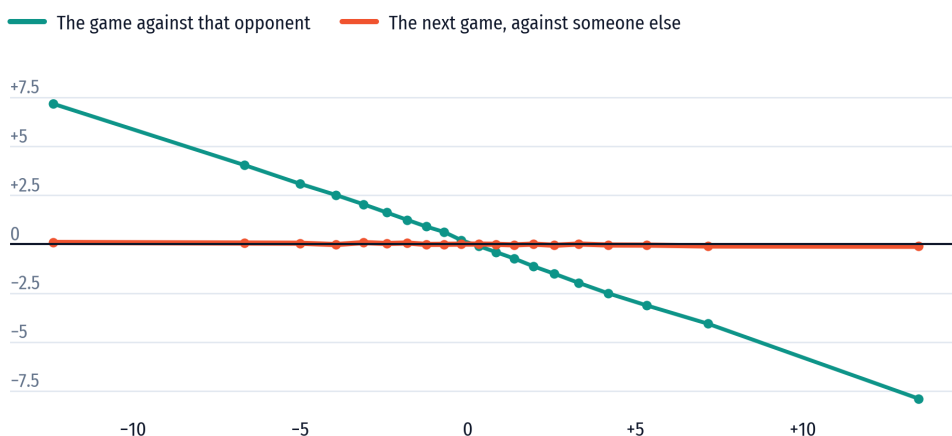
The luck of the draw

To tell the two apart we need players who lost for a reason that had nothing to do with their own form. The pairing system supplies them. Lichess matches players of similar rating, but some opponents are much better than their rating says: they are improving fast, have a new account, or are cheating. Others are worse. Which kind a player meets is luck. We measured every opponent's hidden strength from their results over the month against everyone else, and compared players who drew a secretly strong opponent with players who drew a secretly weak one.

The luck decides a lot in that game. The unluckiest twentieth of players scored 44% in it, and the luckiest twentieth 59%. If losing itself does damage, the unlucky should then do worse in their next game too, against somebody else. They do, slightly.

Bad luck costs one game and a sliver of the next

Score compared with what the ratings predict, percentage points. Across: how much better the first of two opponents in a row was than their rating says, percentage points of score



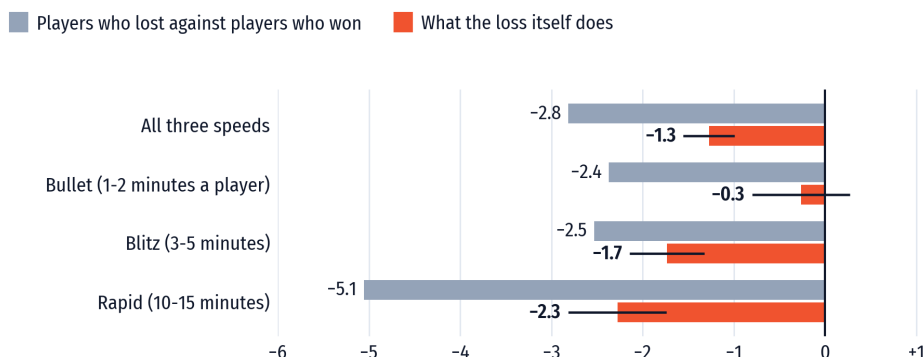
Sources: Lichess open database, rated bullet, blitz and rapid games, September 2026; Signum Average calculations
86m pairs of games in twenty equal groups. Hidden strength: the opponent's results over the month against everyone else

Scaled up to a full loss in place of a win, the damage to the next game is 1.3 percentage points, with a 95% range of 1.0 to 1.6. That is what a player would lose by dropping about 14 rating points. The raw gap between players who lost and players who won, measured the same way, is 2.8 points. So a little under half of it is tilt. (The estimate rests on 86m pairs of games in a row against different opponents who met once in the month. One unit of an opponent's hidden strength lowers the score in the first game by 0.63 and in the second by 0.0081, with a standard error of 0.0009.) The rest would have been there anyway: those players were having a bad evening, or were rated too high, which is why they lost in the first place.

The clock matters. In bullet, where each player has a minute or two for the whole game, a loss costs 0.3 points, too little to tell from nothing. In blitz it costs 1.7. In rapid, where a game can take half an hour, it costs 2.3, as much as dropping 25 rating points. The standard errors are 0.2 to 0.3 points. By rating there is no pattern: the cost is 1.1 points for players under 1200, 1.5 at 1200-1599, 0.9 at 1600-1999 and 1.6 at 2000 and over.

Half form, half tilt

Score in the next game after a loss instead of a win, percentage points, measured against the rating held before the first game



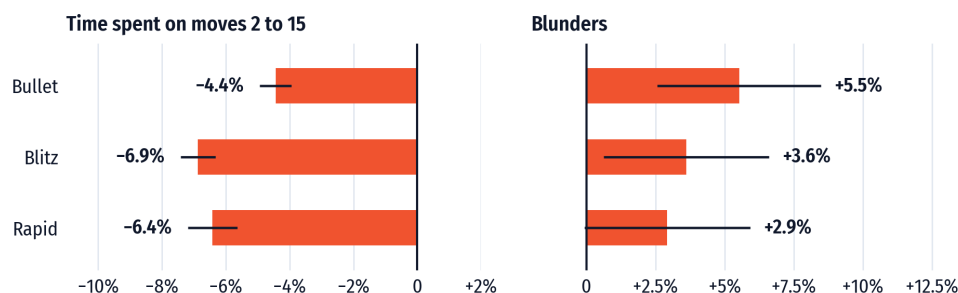
Sources: Lichess open database, rated bullet, blitz and rapid games, September 2026; Signum Average calculations
The thin line is the 95% range. The loss itself is isolated by the luck of the draw

Faster and sloppier

The games themselves show what goes wrong. After a loss, players hurry. They spend 7% less time on moves 2 to 15 of their next game than they would after a win: about three seconds out of 49. And they play worse moves. In the tenth of games that have been run through a chess engine, players who have just lost make 4% more blunders, the moves that throw away a large share of a player's winning chances. In numbers: 0.23 more blunders per 100 moves on an average of 5.9, and 0.14 more points of winning chances thrown away per move on an average of 4.4. Their next game is also nearly two half-moves shorter.

Faster and sloppier

What a loss, instead of a win, does to play in the next game, % change



Sources: Lichess open database, rated bullet, blitz and rapid games, September 2026; Signum Average calculations
The thin line is the 95% range. Blunders are moves Lichess marks ??, in the tenth of games with a computer analysis

Bullet players hurry and blunder after a loss like everyone else. It just costs them too little to show up reliably in their results.

One more game

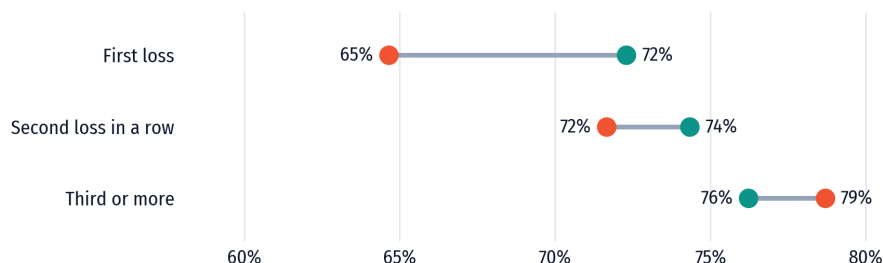
Losing also changes whether people keep playing. By the same test, a loss makes a player 8 points less likely to start another game within half an hour. After an ordinary loss 65% of losers play on, against 72% of the players who beat them.

But the pattern turns as the losses mount. A second loss in a row narrows the gap. A third reverses it. Of players who have just lost three or more in a row, 79% start another game, more than the 76% of the players who beat them. And the loser is in more of a hurry: when both players go on to another game, the loser's starts first 56% of the time, and 63% of the time in rapid. The median wait between the end of a game and the start of the next is 30 seconds for losers and 32 for winners.

One more game

Share of players who start another rated game within 30 minutes, by the loser's run of losses

— Loser — Winner of the same game



Sources: Lichess open database, rated bullet, blitz and rapid games, September 2026; Signum Average calculations
The winner and the loser of the same game are compared, so the length of the game cannot explain the gap

So the advice to walk away after a few losses is sound, for two different reasons. Half of the damage that follows a loss is the loss's own doing, and it comes from rushing. The other half is a bad evening that was already under way. A player counting rating points might also look at the start of the evening. In the first game of a sitting players score 1.4 points less than in the games that follow, as big a handicap as a loss leaves behind.

What this analysis cannot tell you

What happens to players who stop. About a third of losers do not play again within half an hour, and the estimate covers only those who do. Among players who drew a strong opponent, those who had been playing better were a little likelier to play on, which makes our figure, if anything, too small.

How far to trust one method. Our first version of this test gave 0.8 points, not 1.3. A check showed it was flawed: if you have just beaten someone and play them again, their rating has just fallen, so they look secretly stronger than they are. The published figure uses only pairs of players who met once in the month. On those, a test that should give zero (does your next opponent's hidden strength "affect" the game before it?) gives zero. The corrected test has been run on one month. The first version gave much the same answer for August as for September, 1.0 points and 0.8, so the months agree; August has yet to be rerun with the fix.

How bad the blunders are for everyone. Move quality can be measured only in the tenth of games someone chose to analyse, which is not a random tenth.

Why slower games tilt more. That is a pattern, not an experiment. Some secretly strong opponents are cheats, and losing to one may feel different from losing to a fast-improving child. The data are rated games on one site, and leave out classical chess, unrated games and variants.

Appendix: data and methods

Data

The Lichess open database (CC0): every rated standard game of September 2026 (UTC), 89.6m games, and of August 2026, 91.9m, used to repeat the first version of the test. We kept bullet, blitz and rapid games that ended normally, on time or by insufficient material, and dropped games against bots, abandoned games and rule infractions: 87.3m games from 1.9m accounts in September. From each game we used the players, their ratings, the result, the start time, the event, the time control and the clocks, and, where the game carries a computer analysis (about one in ten), the evaluation after each move and the moves Lichess marks as mistakes and blunders. Player names were used only to put a player's games in order and are not published.

Measures

A win scores 1, a draw $\frac{1}{2}$ and a loss 0. The expected score is not the textbook Elo formula but the average score of all games in the same speed, colour and 25-point band of rating gap. Excess score is score minus expected score, in percentage points. A sitting is a run of games by one player whose starts are at most 30 minutes apart. All comparisons use pairs of consecutive games by one player in the same sitting and speed, both ordinary pairings; arena and Swiss tournament games count as part of a player's history but are not tested, because a tournament fixes when and whom a player meets. Near an even game on Lichess, one percentage point of score is worth about 11 rating points (on the textbook Elo formula it would be about 7).

The naive comparison

The first chart sets each result against the rating shown at the start of that game, which already includes the change from the game before. The 2.8-point gap between players who lost and players who won sets the second game against the rating a player held before the first, so that it can be compared with the estimate below.

The luck of the draw

The estimate is an instrumental-variables one. The instrument is the hidden strength of the opponent in the first of two games: that opponent's level over the month (mean excess score in all their other games in that speed, plus their mean rating in those games converted to score) set against the rating they showed in the game. Opponents need at least 20 other games. The treatment is the player's score in the first game. The outcome is the player's excess score in the second, expected score being taken from the rating held before the first game. The comparison holds fixed the speed and the player's rating at the first game, in 25-point bands: among opponents met at the same shown rating, who turns out to be secretly strong is chance. Only pairs in which the player and the first opponent met once in the month are used: 86.0m pairs in September. One unit of hidden strength lowers the score in the first game by 0.63 and the excess score in the second by 0.0081 (standard error 0.0009), which gives 1.27 points per loss in place of a win (standard error 0.14). Standard errors are a jackknife over 50 groups of players.

What was fixed in advance, and what changed

The specification was written before any result was computed, and every change is logged with its date and reason. It changed three times. Twice before the full data were built, after a test on the first 8m games: an opponent's hidden strength is set against the rating they showed in the game, not measured as excess score alone (ratings react to results, so an opponent who loses to you scores above expectation afterwards); and the player's rating is held fixed in 25-point bands, not 200-point ones.

The third change came after the first full results. A balance test, planned in advance, asks whether a player's result in the game *before* the first depends on the first opponent's hidden strength. It cannot, yet it did, strongly: a coefficient of plus 0.017 where the main one was minus 0.005. The cause was rematches. If you have just beaten someone, their rating has just dropped, so against that rating they look secretly strong; if you have just lost to them, weak. That tied the instrument to the player's own form and worked against finding tilt. Among pairs who met once in the month the imbalance is gone (minus 0.0016, standard error 0.0009). The estimate on all pairs was 0.79 points; leaving out only immediate rematches gives 0.87; on pairs who met once, 1.27. All three are in the data files.

The checks

Placebo. The hidden strength of the opponent in the following game cannot affect the current one. On pairs who met once the coefficient is minus 0.0005 (standard error 0.0011), against a main coefficient of minus 0.0081. On all pairs it had been minus 0.0020, two-fifths of that version's main coefficient and a warning we noted at the time.

Who plays on. Among players in the estimate, those who drew a strong opponent had done slightly better in the game before (coefficient plus 0.005, standard error 0.001). This is selection: a loss makes a player 7.7 points less likely to play again within 30 minutes, and those in better form are likelier to continue. It makes the estimate too small, by little, since form carries from one game to the next at only a few per cent.

Time control. Holding the first game's exact time control fixed, not just its speed, gives 1.34 points.

Second month. The first version of the estimate, on all pairs, was repeated unchanged on August 2026 (89.7m games kept, 97.7m pairs). It gives 0.99 points (standard error 0.19) against September's 0.79 (0.14): bullet minus 0.06 (0.27), blitz 1.51 (0.26), rapid 1.97 (0.31), against 0.40 the other way, 1.28 and 2.18 in September. Its placebo was minus 0.0021 (0.0009) in August and minus 0.0020 (0.0011) in September, the same leftover in both, which the rematch fix removes in September. The corrected version has not yet been run on August; this note will be updated when it has.

Splits

By speed: bullet 0.27 points (standard error 0.27), blitz 1.73 (0.21), rapid 2.28 (0.28). By rating: 1.1 points under 1200, 1.5 at 1200-1599, 0.9 at 1600-1999 and 1.6 at 2000 and over, with standard errors of about 0.3: no clear pattern. A split by the gap between the two starts, planned in advance, is in the data file for the first version but should not be read as evidence: that gap is mostly the length of the first game, which depends on both the opponent's strength and the player's own form, so sorting by it manufactures a pattern.

How they play

The same instrument and pairs, with other outcomes in the second game. Time is the clock after a player's 1st move less the clock after the 15th, plus increments, with the first game's time control held fixed: a loss cuts it by 3.3 seconds on a mean of 48.8 (bullet 1.0 of 22.2, blitz 3.8 of 55.2, rapid 7.0 of 108.7). In games with a computer analysis (7.6m pairs), a loss adds 0.23 blunders per 100 moves to a mean of 5.9 (standard error 0.07), 0.55 mistakes and blunders to a mean of 15.9, and 0.14 points of winning chances thrown away per move to a mean of 4.4. Blunders and mistakes are the moves Lichess marks ?? and ?. A game has an analysis when someone asked for one, so these are not a random tenth of games. The next game is also 1.7 half-moves shorter.

Behaviour

The winner and the loser of each decisive ordinary game (77.6m games) are compared, so the length of the game is the same for both. "Plays again" means starting another rated bullet, blitz or rapid game within 30 minutes of the start of that game. Games from the last hour of the month are left out. The median wait between the end of a game and the start of the next is 30 seconds for losers and 32 for winners. The first-game figure compares the first game of every sitting with all later ones, against the rating shown at the start of each: minus 1.0 and plus 0.4 points.

Not done

Classical, unrated and variant games. Other sites. Whether a pause helps: players choose when to pause, and we found no clean way to separate the pause from the player.